Troy Mayer

Mountainside, NJ 07092 | 908-873-9490 | tmayer2112@gmail.com | US Citizen
linkedin.com/in/troyarkin | Portfolio and code: tmayer2112-ui.github.io/Troy_Mayer

EDUCATION

Purdue University, School of Mechanical Engineering and School of Science West Lafayette, IN
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics May 2028
GPA 3.32/4 | Clubs: Club Swimming, House Band

TECHNICAL SKILLS

- **Controls and estimation:** control algorithms (PI, feed-forward, anti-windup, impedance, QP force control), model-based controller tuning, state estimation, Kalman filtering (error-state EKF), sensor fusion, observability analysis
- **Robotics and simulation:** rigid-body kinematics and dynamics, floating-base forward kinematics, MuJoCo physics simulation, Eigen, closed-loop simulation and validation
- **Software:** C++17, Python, MATLAB, SQL, Linux, Git, CMake, GoogleTest unit testing, GitHub Actions CI, real-time embedded (FreeRTOS, ESP32-S3)
- **Data, ML and hardware:** PyTorch, XGBoost, LSTM, Pandas, SciPy, Flask, Plotly, SQLite, Snowflake | KiCad, LabVIEW, DAQ, CATIA, NX, Creo, FEA

EXPERIENCE

Boston Dynamics Waltham, MA
Sensors and Perception Co-Op, Atlas Jul 2026 – Present

- Developing rigid-body kinematics and dynamics for Atlas impacts: floating-base forward kinematics (quaternions, Rodrigues' rotations, velocity composition) from joint encoders maps impacts onto the robot mesh, and Khatib's operational-space effective mass estimates impact energy.
- Designed a delta-v impact-severity metric robust to sensor bandwidth limits, validating its filtering against synthetic pulses of known size and measuring its noise floor; building a physics-based MuJoCo impact simulation.
- Built an impact data-analysis pipeline and Flask/Plotly dashboard on Linux covering ~28k impact events from ~900 robot test logs (telemetry ETL into SQLite, 3D impact heatmap, per-event waveforms), used by engineers during testing.
- Modeling hardware thermal behavior in Python (SciPy), fitting heating curves by multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Caterpillar (via Purdue Data Mine) West Lafayette, IN
Machine Learning Researcher Jan 2026 – May 2026

- Built LSTM and XGBoost machine learning models forecasting next-day paint consumption from five years of data, with Monte Carlo confidence intervals, achieving a 4.7% reduction in material usage; Python data pipelines contributed to a 3.3% cycle-time reduction.

Chortos Lab, Soft Adaptive Systems and Structures West Lafayette, IN
Undergraduate Researcher Aug 2025 – May 2026

- Wrote a Python computer-vision tracker resolving displacement to 0.01 mm and built LabVIEW/MATLAB data acquisition for 50,000+ point datasets, guiding gel formulations that increased force output by over 50%.

ROBOTICS PROJECTS

Legged Robot State Estimation and Control | *Personal Project* Aug 2026 – Oct 2026

- State estimation: built a contact-aided error-state Kalman filter in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a simulated Unitree Go1 in MuJoCo, reaching 0.013 m/s velocity RMSE vs. 1.38 m/s IMU-only, in ~20 µs per real-time update.
- Verification: Jacobians checked against finite differences in GoogleTest unit tests, C++ matched to a Python reference within 1e-13, and observability analysis confirming position and yaw drift while roll, pitch, and velocity stay bounded.
- Control: ran walking and 500 Hz QP force-balance controllers on the estimate with 0 falls in 5 × 60 s walks (QP after a 4 s startup turn); the QP controller cut ground-truth velocity-tracking error from 0.169 to 0.091 m/s.

Active Orthosis, Purdue ASME Biomechanics | *Electronics Team Lead* Apr 2024 – May 2026

- Embedded real-time control: designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3 (FreeRTOS), model-based tuned with feed-forward and anti-windup (0.8% speed error in simulation), with 10 fault monitors, a hardware watchdog, and a kill switch.
- Signal processing: cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian EMG estimator, validated in closed-loop simulation on real recorded EMG with 26 GoogleTest tests in CI.
- Led a 5-engineer team; redesigned the 4-channel EMG board in KiCad (INA333, right-leg drive, ISO7763 isolation), raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot | *Independent Research, Purdue University & UC3M (Madrid)* Oct 2026 – Present

- Motion control and optimization: designing optimized push-off control to hit a target launch speed, knee impedance control for landing, and reaction-wheel attitude control, with MuJoCo simulation comparing risk-limited hop rules.